Technical University of Denmark

Advanced machine learning (02460)

Exam, 15 May 2025

Exam duration: 2 hours.

Aid: Only written works of reference are permitted.

IMPORTANT: Only information written on the pages 2–11 will be evaluated.

Multiple choice: For multiple choice questions, select only one answer.

Info about you: *You must write your student number on all pages at the bottom-left.*

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Part I
Generative models

Exercise A Variational autoencoder (VAE)

Consider a standard deep latent variable model with latent variable $z \in \mathbb{R}$ and observed variable $\mathbf{x} \in \mathbb{R}^2$. We assume that z follows a Mixture of Gaussians (MoG) prior given by

$$p(z) = \frac{1}{2}\mathcal{N}(z \mid -1, 1) + \frac{1}{2}\mathcal{N}(z \mid 3, 1), \tag{1}$$

and the likelihood of $\mathbf{x}$ given z is given by

$$p(\mathbf{x}|z) = \mathcal{N}\left(\mathbf{x} \middle| \rho\left(\begin{bmatrix} -z \\ z \end{bmatrix}\right), \mathbf{I}_2\right), \tag{2}$$

where $\mathbf{I}_2 \in \mathbb{R}^{2\times 2}$ is the identity matrix and $\rho(\mathbf{x}) = \max(0, \mathbf{x})$ is the ReLU activation function. Consider also a variational posterior given by

$$q(z|\mathbf{x}) = \mathcal{N}\left(z \middle| \mu = \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{x}, \sigma^2 = 1\right). \tag{3}$$

Recall that the ELBO for a VAE is given by

$$\mathrm{ELBO}(\mathbf{x}) = \mathbb{E}_{z\sim q(z|\mathbf{x})}\left[\ln \frac{p(\mathbf{x} \mid z)p(z)}{q(z \mid \mathbf{x})}\right] = \mathbb{E}_{z\sim q(z|\mathbf{x})}[\ln p(\mathbf{x} \mid \mathbf{z})] - \mathrm{KL}[q(z \mid \mathbf{x}) \,\|\, p(z)]. \tag{4}$$

Also, recall that the densities of a univariate and a bivariate Gaussian with identity covariance are given by

$$\mathcal{N}(z \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \quad \text{and} \quad \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{I}_2) = \frac{1}{2\pi} e^{-\frac{1}{2}\|\mathbf{x}-\boldsymbol{\mu}\|^2}. \tag{5}$$

Question A.1: Assuming a one-sample Monte Carlo estimate of ELBO, where the sampled value is $z' = 1$, what is the ELBO evaluated at $\mathbf{x}' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$?

Exercise B Normalizing flows

Consider a normalising flow with random variables $\mathbf{u}, \mathbf{x} \in \mathbb{R}^2$, where the base distribution is given by

$$p_{\mathrm{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u}|\mathbf{0}, \mathbf{I}_2) = \frac{1}{2\pi} e^{-\frac{1}{2}(u_1^2+u_2^2)}, \tag{6}$$

where $\mathbf{I}_2 \in \mathbb{R}^{2\times 2}$ is the identity matrix, and the density of $\mathbf{x}$ is given by

$$p_{\mathrm{x}}(\mathbf{x}) = p_{\mathrm{u}}(\mathbf{u})|\det J_T(\mathbf{u})|^{-1} \quad \text{where} \quad \mathbf{u} = T^{-1}(\mathbf{x}). \tag{7}$$

Consider the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$T(\mathbf{u}) = \begin{bmatrix} u_1 \\ 2u_2 + f(u_1) \end{bmatrix}, \tag{8}$$

where $f : \mathbb{R} \to \mathbb{R}$ is a continuously differentiable (C^1) scalar function.

Question B.1: Which of the following correctly describes **all functions** $f \in C^1$ for which T is invertible?

A ☐ f **must** be linear

B ☐ f **must** be continuously differentiable and injective

C ☐ f **must** be continuously differentiable and bijective

D ☐ f can be any continuously differentiable function

Give a brief justification.

Question B.2: What is the density $p_{\mathrm{x}}(\mathbf{x})$ at $\mathbf{x} = [1, 3]^\intercal$, assuming $f(u_1) = u_1$? Give a brief justification.

Exercise C Denoising Diffusion Probabilistic Models (DDPM)

Consider a DDPM model of real-valued data with $T = 5$ steps, where $\mathbf{x}_0 \in \mathbb{R}^D$ is that data variable and $\mathbf{x}_1, \ldots, \mathbf{x}_5 \in \mathbb{R}^D$ are the latent variables. Recall that the forward process in the DDPM is a Markov chain that gradually adds Gaussian noise to the data using, i.e.,

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}), \tag{9}$$

where $\{\beta_t\}_{t=1}^{T}$ is the variance schedule.

Question C.1: Which one of the following equalities is true for any variance schedule?

A ☐ $q(\mathbf{x}_2 \mid \mathbf{x}_3, \mathbf{x}_0, \mathbf{x}_5) = q(\mathbf{x}_2 \mid \mathbf{x}_0)$

B ☐ $q(\mathbf{x}_2 \mid \mathbf{x}_3, \mathbf{x}_0, \mathbf{x}_5) = q(\mathbf{x}_2 \mid \mathbf{x}_5)$

C ☐ $q(\mathbf{x}_2 \mid \mathbf{x}_3, \mathbf{x}_0, \mathbf{x}_5) = q(\mathbf{x}_2 \mid \mathbf{x}_3, \mathbf{x}_0)$

D ☐ $q(\mathbf{x}_2 \mid \mathbf{x}_3, \mathbf{x}_0, \mathbf{x}_5) = q(\mathbf{x}_2 \mid \mathbf{x}_3, \mathbf{x}_5)$

Give a brief justification for why the chosen equality holds.

Part II

Representation learning and identifiability

Exercise D Pullback metrics

Consider the generative model

$$f(\mathbf{x}) = \mathbf{W}_2 \tanh\left(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1\right) + \mathbf{b}_2, \tag{10}$$

where $\mathbf{W}_1 \in \mathbb{R}^{D\times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{D\times D}$ are full-rank matrices ($D \geq d$), and $\mathbf{b}_1 \in \mathbb{R}^D$ and $\mathbf{b}_2 \in \mathbb{R}^D$ are bias vectors. Here tanh is used as the (elementwise) activation function.

Recall that the derivative of tanh is

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh(x) = 1 - \tanh^2(x). \tag{11}$$

Let $\mathrm{diag} : \mathbb{R}^D \to \mathbb{R}^{D\times D}$ denote the function that produces a diagonal matrix from a vector. The pull-back metric $\mathbf{G}_\mathbf{x} = \mathbf{J}_\mathbf{x}^\intercal\mathbf{J}_\mathbf{x}$, where $\mathbf{J}_\mathbf{x}$ is the Jacobian of f, is then one of the following:

(A) $\mathbf{G}_\mathbf{x}^{(A)} = \mathbf{W}_1^\intercal\mathbf{W}_2^\intercal\mathbf{W}_2\mathbf{W}_1 + \mathbf{W}_1^\intercal\Lambda_1\mathbf{W}_2^\intercal\mathbf{W}_2\Lambda_1\mathbf{W}_1 - \mathbf{W}_1^\intercal\mathbf{W}_2^\intercal\mathbf{W}_2\Lambda_1\mathbf{W}_1 - \mathbf{W}_1^\intercal\Lambda_1\mathbf{W}_2^\intercal\mathbf{W}_2\mathbf{W}_1$

(B) $\mathbf{G}_\mathbf{x}^{(B)} = \mathbf{W}_2\mathbf{W}_1\mathbf{W}_1^\intercal\mathbf{W}_2^\intercal + \mathbf{W}_2\Lambda_1\mathbf{W}_1\mathbf{W}_1^\intercal\Lambda_1\mathbf{W}_2^\intercal - \mathbf{W}_2\mathbf{W}_1\mathbf{W}_1^\intercal\Lambda_1\mathbf{W}_2^\intercal - \mathbf{W}_2\Lambda_1\mathbf{W}_1\mathbf{W}_1^\intercal\mathbf{W}_2^\intercal,$

(C) $\mathbf{G}_\mathbf{x}^{(C)} = \mathbf{W}_1^\intercal\mathbf{W}_2^\intercal\mathbf{W}_2\mathbf{W}_1 + \mathbf{W}_1^\intercal\Lambda_2\mathbf{W}_2^\intercal\mathbf{W}_2\Lambda_2\mathbf{W}_1 - \mathbf{W}_1^\intercal\mathbf{W}_2^\intercal\mathbf{W}_2\Lambda_2\mathbf{W}_1 - \mathbf{W}_1^\intercal\Lambda_2\mathbf{W}_2^\intercal\mathbf{W}_2\mathbf{W}_1$

(D) $\mathbf{G}_\mathbf{x}^{(D)} = \mathbf{W}_2\mathbf{W}_1\mathbf{W}_1^\intercal\mathbf{W}_2^\intercal + \mathbf{W}_2\Lambda_2\mathbf{W}_1\mathbf{W}_1^\intercal\Lambda_2\mathbf{W}_2^\intercal - \mathbf{W}_2\mathbf{W}_1\mathbf{W}_1^\intercal\Lambda_2\mathbf{W}_2^\intercal - \mathbf{W}_2\Lambda_2\mathbf{W}_1\mathbf{W}_1^\intercal\mathbf{W}_2^\intercal,$

where

$$\Lambda_1 = \mathbf{I} - \mathrm{diag}\left(\tanh^2(\mathbf{W}_1\mathbf{x})\right) \tag{12}$$

$$\Lambda_2 = \mathbf{I} - \mathrm{diag}\left(\tanh^2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\right) \tag{13}$$

Question D.1: Which one of the above four options is correct?

A ☐ The pull-back metric is $\mathbf{G}_\mathbf{x} = \mathbf{G}_\mathbf{x}^{(A)}$.

B ☐ The pull-back metric is $\mathbf{G}_\mathbf{x} = \mathbf{G}_\mathbf{x}^{(B)}$.

C ☐ The pull-back metric is $\mathbf{G}_\mathbf{x} = \mathbf{G}_\mathbf{x}^{(C)}$.

D ☐ The pull-back metric is $\mathbf{G}_\mathbf{x} = \mathbf{G}_\mathbf{x}^{(D)}$.

Exercise E Numerical computation of geodesics

Consider a generative model $\mathbf{y} = f(\mathbf{x})$, where $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathbb{R}^D$ with $d < D$. Assume that $f : \mathbb{R}^d \to \mathbb{R}^D$ is a smooth nonlinear function.

We now set out to numerically estimate the shortest path (geodesic) between latent points $c_0 \in \mathbb{R}^d$ and $c_{N+1} \in \mathbb{R}^d$ by constructing a piecewise linear latent path with nodes $c_{1:N} = \{c_1, \ldots, c_N\}$. Consider the length-minimizing path on the manifold spanned by f

$$c^*_{1:N} = \arg\min_{c_{1:N}} \sum_{n=1}^{N+1} \|f(c_n) - f(c_{n-1})\| \,. \tag{14}$$

Question E.1: Is the piecewise linear path with nodes $\{c_0, \ldots, c_{N+1}\}$ a discrete approximation to the shortest path connecting c_0 with c_{N+1}?

A ☐ Yes.

B ☐ No.

Explain your reasoning:

E

Exercise F Curve energies

Consider the generative model $\mathbf{y} = f(\mathbf{x})$, where

$$f(\mathbf{x}) = \begin{pmatrix} 2x_1^2 \\ x_1^2 + 3x_2^2 \\ 2x_1 + x_2^2 \end{pmatrix}. \tag{15}$$

Furthermore, consider the two latent curves

$$c_1(t) = t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad c_2(t) = t \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{16}$$

Question F.1: Derive the Riemannian pull-back metric $\mathbf{G}_{\mathbf{x}} = \mathbf{J}_{\mathbf{x}}^{\intercal}\mathbf{J}_{\mathbf{x}}$, where $\mathbf{J}_{\mathbf{x}}$ denotes the Jacobian of f at $\mathbf{x}$.

Question F.2: Let the energy of a curve be given by

$$\mathcal{E}(c) = \int_0^1 \dot{c}(t)^{\intercal}\mathbf{G}_{c(t)}\dot{c}(t)\mathrm{d}t, \tag{17}$$

where $\dot{c}(t) = \frac{\mathrm{d}c(t)}{\mathrm{d}t}$ is the curve derivative. Evaluate the energy of the curves c_1 and c_2.

Exercise G Reparametrization of neural networks

Consider the neural network $f : \mathbb{R} \to \mathbb{R}$

$$f(x|w_1, w_2) = w_1 \sigma(w_2\, x) \qquad\qquad x,\, w_1,\, w_2 \in \mathbb{R}, \tag{18}$$

where σ is the 'parametric ReLU' activation function defined as

$$\sigma(x) = \begin{cases} \alpha\, x, & x < 0 \\ x, & x \geq 0 \end{cases}, \qquad\qquad \alpha \in \mathbb{R}, \tag{19}$$

where α is a priori specified, i.e., it is not considered a learnable parameter. Further consider another set of parameters (w_3, w_4) for the neural network.

Question G.1: Let $w_3 = \beta\, w_1$ for $\beta \neq 0$. Derive a value for w_4 such that

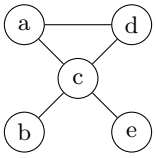
$$f(x|w_1, w_2) = f(x|\beta\, w_1, w_4), \qquad \text{for all } \beta \neq 0. \tag{20}$$

Question G.2: Discuss the implications of your answer to the previous question.

Part III
Graph models

Exercise H Metrics

Consider the following graph



The clustering coefficient is defined as:

$$c_u = \frac{|(v_1, v_2) \in \mathcal{E} : v_1, v_2 \in \mathcal{N}(u)|}{\binom{d_u}{2}}$$

Question H.1: What is the clustering coefficient of node a and node c?

$c_{\mathrm{a}} =$ _____, $c_{\mathrm{c}} =$ _____
Show the main steps of your derivation.

The eigenvector centrality is defined as:

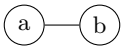
$$e_u = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} e_v$$

Question H.2:
The eigenvector centralities of node a, b, d, and e are $e_{\mathrm{a}} = e_{\mathrm{d}} = 0.473$, and $e_{\mathrm{b}} = e_{\mathrm{e}} = 0.271$. What is the eigenvector centrality of node c?

$e_{\mathrm{c}} =$ _____
Show the main steps of your derivation.

Exercise I Graph Fourier transform

Consider the following graph with two nodes:



The eigenvalues of the adjacency matrix are given by $\lambda_1 = 1$ and $\lambda_2 = -1$.

A graph convolution with a finite-length filter can be written in the vertex domain, and in the spectral domain as:

$$\boldsymbol{y} = \sum_{k=0}^{N} h[k]\boldsymbol{A}^k\boldsymbol{x}, \qquad \tilde{\boldsymbol{y}} = \sum_{k=0}^{N} h[k]\boldsymbol{\Lambda}^k\tilde{\boldsymbol{x}}. \tag{21}$$

We will use a filter given by:

$$h[k] = \begin{cases} 1 & 0 \leq k \leq 4 \\ 0 & \text{otherwise.} \end{cases}$$

Question I.1: Given a vertex-domain signal $\boldsymbol{x} = [1, 2]^\top$, what is the filtered vertex-domain signal $\boldsymbol{y}$?

$\boldsymbol{y} = [_____, _____]^\top$

Show the main steps of your derivation.

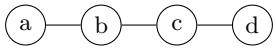
Question I.2: Given a spectral-domain signal $\tilde{\boldsymbol{x}} = [2, 3]^\top$, what is the filtered spectral-domain signal $\tilde{\boldsymbol{y}}$?

$\tilde{\boldsymbol{y}} = [_____, _____]^\top$

Show the main steps of your derivation.

Exercise J Message passing

Consider the following graph:



We will use a simple message-passing graph neural network (GNN) with *two layers*, defined as

$$h_u^{(k)} = w_0 + w_1 h_u^{(k-1)} + w_2 \sum_{v \in \mathcal{N}(u)} h_v^{(k-1)}, \tag{22}$$

with parameters $w_0 = w_1 = w_2 = 1$. Each node has initial scalar features $h_u^{(0)} = 1$.

Question J.1: What are the updated node features after the two message-passing rounds?

$h_{\mathrm{a}}^{(2)} =$ _____, $h_{\mathrm{b}}^{(2)} =$ _____, $h_{\mathrm{c}}^{(2)} =$ _____, $h_{\mathrm{d}}^{(2)} =$ _____

Show the main steps of your derivation.

Question J.2: We can write a single layer of the GNN as a matrix vector product

$$\boldsymbol{h}^{(k)} = \boldsymbol{w}_0 + \boldsymbol{M}\boldsymbol{h}^{(k-1)}, \tag{23}$$

where $\boldsymbol{h}^{(k)} = [h_{\mathrm{a}}^{(k)}, h_{\mathrm{b}}^{(k)}, h_{\mathrm{c}}^{(k)}, h_{\mathrm{d}}^{(k)}]^\top$ and $\boldsymbol{w}_0 = [w_0, w_0, w_0, w_0]^\top$. Again using parameters $w_0 = w_1 = w_2 = 1$, fill in all the non-zero values of the matrix $\boldsymbol{M}$.

$$M = \begin{bmatrix} __, __, __, __ \\ __, __, __, __ \\ __, __, __, __ \\ __, __, __, __ \end{bmatrix}$$

Question J.3: How can we choose the parameters, such that the GNN implements the identity function?

$w_0 =$ _____, $w_1 =$ _____, $w_2 =$ _____.